

## Quiz 9: Joint Distributions, Convolutions, and Conditioning

1. (2 points) **Conditional distribution of a centered bivariate normal.**

Let  $(X_1, X_2)$  have joint density

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x_1^2 + x_2^2 - 2\rho x_1 x_2}{2(1-\rho^2)}\right), \quad -1 < \rho < 1.$$

Show that for every real number  $a$ ,

$$X_1 \mid X_2 = a \sim \mathcal{N}(\rho a, 1 - \rho^2).$$

2. (2 points) **Poisson thinning.**

Let  $N \sim \text{Poisson}(\lambda)$  be the number of events that occur. Each event is then independently kept with probability  $p$ , where  $0 < p < 1$ . Let  $X$  denote the number of events that are kept. Show that  $X \sim \text{Poisson}(\lambda p)$ .

3. (2 points) **Infinite divisibility of the Poisson distribution.**

Let  $n \geq 1$  be an integer and let  $X_1, \dots, X_n$  be independent random variables with  $X_i \sim \text{Poisson}(\lambda/n)$ . Use the convolution formula for sums of independent discrete random variables to show that

$$X_1 + \dots + X_n \sim \text{Poisson}(\lambda).$$

4. (2 points) **Infinite divisibility of the normal distribution.**

Let  $n \geq 1$  be an integer and let  $Z_1, \dots, Z_n$  be independent random variables with  $Z_i \sim \mathcal{N}\left(\frac{\mu}{n}, \frac{\sigma^2}{n}\right)$ . Use the convolution formula for sums of independent continuous random variables to show that

$$Z_1 + \dots + Z_n \sim \mathcal{N}(\mu, \sigma^2).$$

5. (2 points) **Knuth's algorithm for generating Poisson random variables.**

Let  $\lambda > 0$  be fixed. Let  $U_1, U_2, \dots$  be independent  $\text{Unif}(0, 1)$  random variables, and define

$$N = \min \left\{ k \geq 0 : \prod_{i=1}^{k+1} U_i \leq e^{-\lambda} \right\}.$$

Show that  $N \sim \text{Poisson}(\lambda)$ .